COMSATS University Islamabad, Lahore Campus

Operating System Assignment #01

Submitted to: Ma'am Zeenat Afzal

Submitted by: Samiullah Ashfaq
Roll# SP24-BSE-107-A

head:
Displays the first part of a file (by default first 10 lines).
Used to quickly preview the beginning of large files.

```
/OS_LabAs_01                                                    57s311ms
head example.txt
Hello Kaizuk!
How are you?
How is your day going?
Take Care
Good-Bye!

/OS_LabAs_01                                                    4ms
head -2 example.txt
Hello Kaizuk!
How are you?

/OS_LabAs_01                                                    4ms
```

Tail:
Displays the last part of a file (by default last 10 lines).
Often used to monitor logs in real time.

```
/OS_LabAs_01                                                    4ms
tail example.txt
Hello Kaizuk!
How are you?
How is your day going?
Take Care
Good-Bye!

/OS_LabAs_01                                                    7ms
tail -3 example.txt
How is your day going?
Take Care
Good-Bye!

/OS_LabAs_01                                                    4ms
```

grep:
Searches for a specific pattern or text inside files.
Prints lines that match the given pattern.

```
/OS_LabAs_01                                                    4ms
grep "G" example.txt
Good-Bye!

/OS_LabAs_01                                                    4ms
grep "H" example.txt
Hello Kaizuk!
How are you?
How is your day going?

/OS_LabAs_01                                                    4ms
```

chmod:
Changes file or directory permissions.
Controls read, write, and execute access.

```
/OS_LabAs_01                                                    4ms
ls -l
total 4
-rw-r--r-- 1 sami sami 70 Feb 22 08:15 example.txt

/OS_LabAs_01                                                    4ms
chmod u+x example.txt

/OS_LabAs_01                                                    4ms
ls -l
total 4
-rwxr--r-- 1 sami sami 70 Feb 22 08:15 example.txt

/OS_LabAs_01                                                    5ms
```

useradd:
Creates a new user account on the system.
Sets up basic user information.

```
/OS_LabAs_01                                                    5ms
useradd agha
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.

/OS_LabAs_01                                                    12ms
sudo useradd agha
[sudo] password for sami:
```

```
/OS_LabAs_01                                                    4ms
grep "home" /etc/passwd
sami:x:1000:1001:Samiullah Ashfaq:/home/sami:/bin/bash
agha:x:1001:1002::/home/agha:/bin/bash

/OS_LabAs_01                                                    4ms
```

passwd:
Sets or updates a user's password.
Ensures secure authentication for accounts.

```
/OS_LabAs_01                                                    12s403ms
sudo passwd agha
New password:
Retype new password:
passwd: password updated successfully
```

find:
Searches for files and directories in a specified path.
Filters results based on name, size, type, or other criteria.

```
/OS_LabAs_01                                                    4ms
ls
example.txt  hello.c  kaizuk.java

/OS_LabAs_01                                                    4ms
find -name "example.txt"
./example.txt

/OS_LabAs_01                                                    4ms
```

Diff:
Compares two files line by line.
Shows the differences between them.

```
/OS_LabAs_01                                                    11s853ms
cat example.txt
Hello Kaizuk!
How are you?
How is your day going?
Take Care
Good-Bye!

/OS_LabAs_01                                                    4ms
cat example2.txt
Hello Kaizuk!
How are you?
Good-Bye!

/OS_LabAs_01                                                    4ms
diff example.txt example2.txt
3,4d2
< How is your day going?
< Take Care

/OS_LabAs_01                                                    8ms
```

gdb:
A debugger used to test and troubleshoot programs.
Helps analyze crashes and inspect program execution.

```
/OS_LabAs_01                                                    8ms
cat > hello.c
#include <stdio.h>
int main() {
  printf("Hello Kaizuk");
}

/OS_LabAs_01                                                    58s107ms
ls
example2.txt  example.txt  hello.c

/OS_LabAs_01                                                    4ms
gcc hello.c

/OS_LabAs_01                                                    41ms
ls
a.out  example2.txt  example.txt  hello.c

/OS_LabAs_01                                                    5ms
./a.out
Hello Kaizuk
/OS_LabAs_01                                                    4ms
gdb ./a.out
```

```
/OS_LabAs_01                                                    1m8s747ms
gdb ./a.out
GNU gdb (Fedora Linux) 17.1-1.fc43
Copyright (C) 2025 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-redhat-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./a.out...

This GDB supports auto-downloading debuginfo from the following URLs:
  <ima:enforcing>
  <https://debuginfod.fedoraproject.org/>
  <ima:ignore>
Enable debuginfod for this session? (y or [n])
```

```
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
(No debugging symbols found in ./a.out)
(gdb)
```